VERSION
14.0

AMS GRA ARCHITECTURE VOLUMES

Agile Mission Suite (AMS) Government Reference Architecture (GRA) Service MPU

AFLCMC/WA

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1.0 Overview

An Agile Mission Suite (AMS) Government Reference Architecture (GRA) Service is a Level 1 Minimum Procurable Unit (MPU) in the AMS GRA. AMS GRA Services provide computing for the Multi-Function Processor (MFP) layer. These Services are deployed on Digital Payload Processors (DPPs) in the Compute Cluster Farm (CCF).

1.1 Purpose

The AMS GRA Service MPU Volume introduces the AMS GRA Service MPU and describes its key features, logical context, interfaces, and compliance requirements. The key features provide design guidance on what is typically expected for the AMS GRA Service MPU. The interface diagram shown in **Figure 2.2.1** contains a graphical representation of the types of interfaces that can exist (from the perspective of the AMS GRA Service MPU). The Interface Table (**Table 2.3.1**) contains a textual representation of the interfaces and their descriptions. Compliance requirements (**Table 3.0.1**) describe what is required for the AMS GRA Service MPU to achieve an AMS GRA compliance tier.

1.2 Scope

The AMS GRA Service MPU Volume supports compliance assessment for AMS GRA adopters.

1.3 Authoritative References

Authoritative references are key documents that a procuring authority, Mission System Integrator (MSI), or MPU provider refers to when determining whether a given AMS GRA entity is compliant with AMS GRA business and architecture requirements. An AMS GRA entity is compliant when it provides the required prescriptive constructs (e.g. interfaces for an MPU). See **Table 1.3.1** for a list of authoritative references for the AMS GRA Service MPU Volume.

Standard	Title	Version/Date
AMS GRA Volumes	AMS GRA Service Contract Template	14.0
AMS GRA Volumes	IR MEL IDD	14.0
AMS GRA Volumes	RF MEL IDD	14.0
AMS GRA Volumes	CDS IDD	14.0
AMS GRA Volumes	Compliance Assessment Approach Volume	14.0

Standard	Title	Version/Date
AMS GRA Volumes	Mission System Volume	14.0

Table 1.3.1: Authoritative References Identify the Key Documents Necessary for Performing a Compliance Assessment of AMS GRA Service MPUs.

1.4 Informative References

Informative references are supplemental documents that a procuring authority, MSI, or MPU provider refers to for more information about an AMS GRA entity or related entities. See **Table 1.4.1** for a list of informative references for the AMS GRA Service MPU Volume.

Standard	Title	Version/Date
AMS GRA Volumes	Executive Summary	14.0
AMS GRA Volumes	Software Architecture Volume	14.0

Table 1.4.1: Informative References Identify Documents a Procuring Authority, MSI, or MPU Provider Can Use for More Information About the AMS GRA Service MPU and Related Entities.

2.0 AMS GRA Service

The AMS GRA Service is the centerpiece of AMS GRA. AMS GRA Services provide a standardized approach for high-performance software deployed on sensing platforms with Open, high-speed interfaces to digital aperture technologies. AMS GRA Services can be mission agnostic or mission specific. The AMS GRA Service is complementary to an Open Mission Systems (OMS) Service and not proposed as a replacement. The AMS GRA Service should be used in situations where an OMS Service cannot be implemented. A service that meets the definition and compliance requirements of OMS and uses only the Abstract Service Bus (ASB) is an OMS Service. An AMS GRA Service meets the definition and compliance requirements of AMS GRA and uses the Avionics Backplane Bus (ABB). An AMS GRA Service may also use the ASB.

AMS GRA Services are built for a DPP Compute Environment, which includes DPP Application Programming Interfaces (APIs), for access to various facilities provided by the DPP and its operating system/environment, such as access to time. AMS GRA Services are composed of one or more Digital Payloads (DPs) that target General Purpose Processor (GPP), Graphics Processing Unit (GPU), or Field-Programmable Gate Array (FPGA) components within a DPP.

AMS GRA Services are required to use Open interfaces to communicate to other AMS GRA Services over the ABB. These open interfaces allow Services the flexibility needed to meet performance requirements and computing resource constraints that preclude the use of the OMS Critical Abstraction Layer. Open interface definitions between Services enable replaceability, competition, and reduced integration cost while preserving the opportunity to rapidly adapt interfaces to emerging capabilities. AMS GRA Services may also use OMS Critical Abstraction Layer Messages on the ASB to communicate to other MPUs such as OMS Services.

AMS GRA Services use the Multi-Function Aperture (MFA) Encapsulation Layer (MEL) interface to communicate over the ABB with MFAs. AMS GRA provides a common definition of both the command/control and the data components to promote interoperability between MFA implementations.

An AMS GRA Service is compliant with prescribed requirements, standards, and interfaces that increase its portability across multiple weapons systems. The AMS GRA Service lessens the impact of upgrades and the replacement of software by tightly defining the boundaries of the service and describing the service functionality sufficiently for third parties to replicate and innovate on those functions.

2.1 Key Features

Table 2.1.1 identifies the key features of AMS GRA Services and the benefit of these features. This table is not prescriptive but identifies a number of attributes that AMS GRA definitions and interfaces are designed to support. This table enables AMS GRA Service providers to summarize the specific features supported by an AMS GRA Service.

Key Feature	Benefit
Automated Build and Testing Pipeline	Provides repeatable, traceable, and efficient creation and testing of digital payloads for AMS GRA Services.
Leveraged OCI-compliant Containers	Standardizes deployed software environment for compatibility with multiple computer hardware instantiations.
Linked MEL Implementation Library	Supports implementation of MEL API commands and data transfers between Services and MFAs.
Open Inter-Service ABB Interface	Avoids limitations of the OMS Critical Abstraction Layer when needed to meet functional requirements.
Standardized Marketplace	Provides standardized application store to pull Services per adopter and platform needs.

Table 2.1.1: The AMS GRA Service Key Features Table Describes the Specific Functionality Supported by an AMS GRA Service.

2.2 Interface Diagram

Figure 2.2.1 depicts the AMS GRA Service and its execution environment provided by the DPP MPU logical interfaces. Interface connections denoted as pass-through only represent situations where the sending MPU is not the original source and does not use or modify what is provided to the receiving MPU.

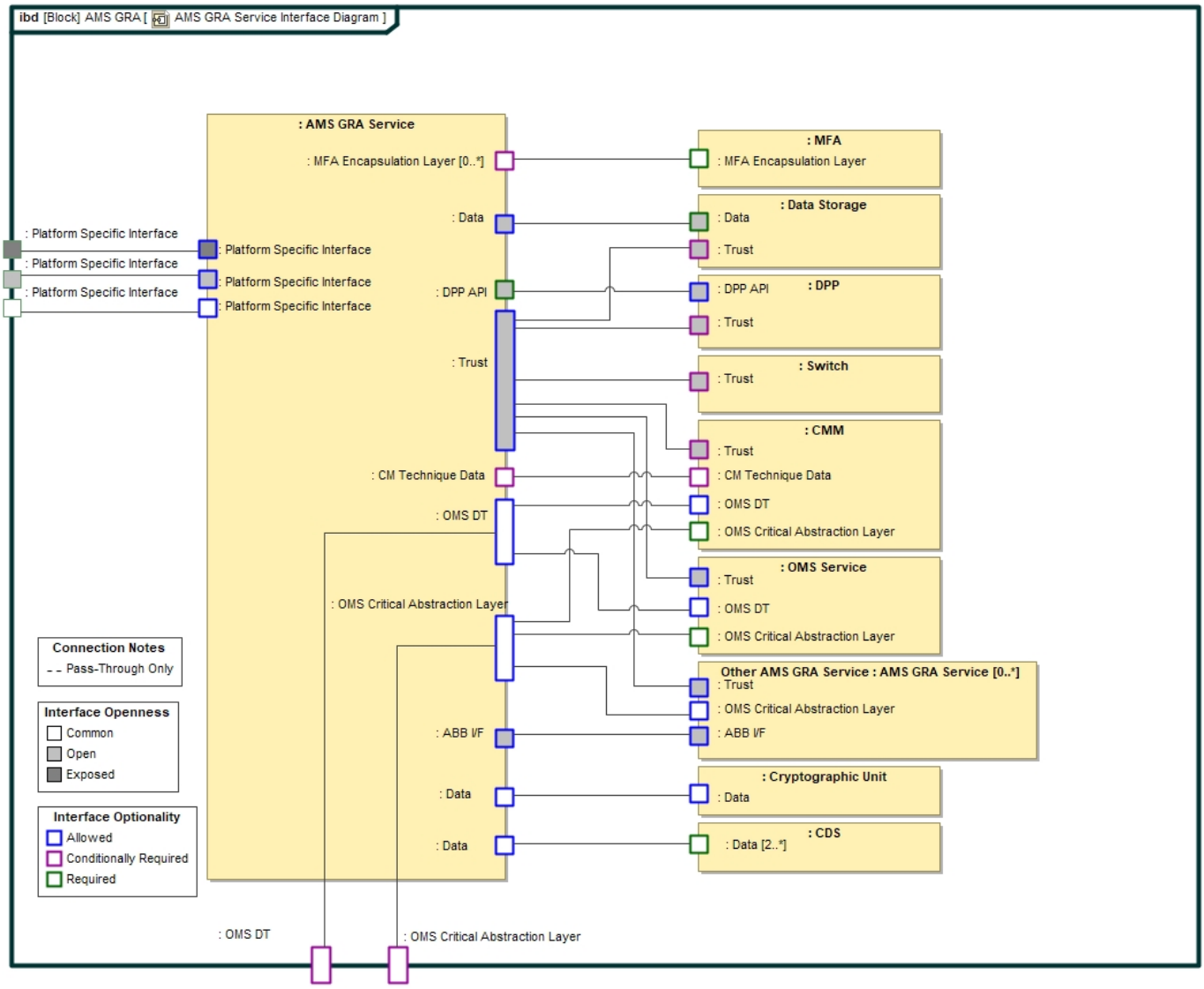


Figure 2.2.1: The AMS GRA Service MPU Interface Diagram Shows Logical Interfaces of the AMS GRA Service and Their Connections to a DPP Execution Environment.

2.3 Interface Table

Table 2.3.1 identifies the AMS GRA Service interfaces depicted in **Figure 2.2.1**. When determining whether a given MPU or infrastructure is AMS GRA compliant, Interface Optionality is defined as follows:

- Required interfaces must be implemented.
- Conditionally Required interfaces must be implemented when specific conditions are satisfied.
- Allowed interfaces are optional to implement.

For interfaces listed multiple times, each instance represents an openness category that an interface could belong to. The Platform Specific Interface can fall into any of the three openness categories because some MPUs may instantiate an allowable interface at one or more openness levels. Those interfaces will be represented on the corresponding diagram and table for each allowable openness level.

Interface	Documentation	Interface Optionality	Condition
Trust	The Trust interface enables secure connectivity between endpoints using cryptographic means to establish a secure channel. This channel may be used to verify integrity, authenticate endpoints, or to transfer data requiring higher levels of assurance than normal data transfers, such as CPI, or Software. The implementation of this interface may vary depending on assurance level, exportability, and the various mission specific requirements of the adopting program.	Allowed	N/A
Platform Specific Interface	The Common Platform Specific Interface provides an information flow between AMS GRA Services and proprietary hardware and software.	Allowed	N/A
Platform Specific Interface	The Exposed Platform Specific Interface provides an information flow between AMS GRA Service and non-proprietary hardware and software that follow AMS GRA standards.	Allowed	N/A
Platform Specific Interface	The Open Platform Specific Interface provides an information flow between AMS GRA Service and non-proprietary hardware and software not required to follow AMS GRA standards.	Allowed	N/A
OMS DT	The OMS Data Transfer interface provides access to the OMS Internet Protocol- based messages when appropriate.	Allowed	N/A

Interface	Documentation	Interface Optionality	Condition
OMS Critical Abstraction Layer	The OMS Critical Abstraction Layer provides access to the UCI message schema. See the OMS standard for additional information.	Allowed	N/A
DPP API	The DPP API interface provides Level 1 MPUs with access to various facilities provided by the DPP Compute Environment. Note that the GPP variant of the DPP covers function calls for services to be able to query for Time as distributed by the Mission System.	Required	N/A
Data	The Data interface specifies the data interface between the AMS GRA Service and the Data Storage.	Allowed	N/A
Data	The SM interface provides connected MPUs with access to SM functionality.	Allowed	N/A
CM Technique Data	The CM Common Data Format interface specifies the VITA-based data format for transferring of countermeasures technique data, which may be generated by an AMS GRA Service, over the ABB.	Conditionally Required	Required for an AMS GRA Service that is performing countermeasures technique generation.
ABB I/F	The ABB interface supports the exchange of large and/or high-speed data between AMS GRA Services.	Allowed	N/A
ABB I/F	The MEL defines the command, control, and data exchange patterns of interaction associated with any AMS GRA compliant MFA. RF MEL details are in the RF MEL IDD. IR MEL details are in the IR MEL IDD.	Conditionally Required	Required for an AMS GRA Service that communicates with an MFA.

Table 2.3.1: The Procuring Authority, MPU Provider, and MSI Use this Table to Identify the Interfaces to Implement for a Compliant MPU or Infrastructure.

3.0 Compliance Requirements

Table 3.0.1 states the applicable MPU compliance requirements for the AMS GRA Service MPU.

See the Compliance Assessment Approach Volume for more information on the tiered requirement strategy as well as the definitions of the interface characteristics.

See the Mission System Volume for additional details about MPU Fault Reporting.

MPU developers need to coordinate with the Procuring Authority and applicable MSI to receive additional guidance beyond basic compliance. The type of additional guidance MPU developers will receive includes architectural guidance to ensure the MPU integrates within the overall Mission System architecture as well as guidance to align with Commercial-Off-The-Shelf (COTS) standards (e.g., physical form factor, cooling strategy, environmental operating conditions, etc.).

Note: The verification methods listed in the requirements table are AMS GRA recommendations. An adopting program may alter the recommended verification methods In Accordance With (IAW) the needs of the program. Refer to the Compliance Assessment Approach Volume for more information.

Id	Text	Compli-ance Tier	Recommended Verification Method	Notes
GRA-MPU-001	An MPU shall be documented IAW the corresponding AMS GRA compliance documentation template or contract template, as applicable.	Tier 1	Inspection	
GRA-MPU-003	An MPU targeting deployment on a GPP DPP shall be restricted to using the POSIX API for OS system calls.	Tier 1	Inspection	
GRA-MPU-005	An MPU with CPI on a Trusted Ops platform shall cryptographically protect CPI using an open implementation.	Tier 3	Analysis	
GRA-MPU-006	An MPU that processes or decrypts CPI on a built-for-export platform shall implement the OMS Standard.	Tier 2	Analysis	
GRA-MPU-019	An MPU that implements an OMS interface shall implement the interface IAW OMS requirements.	Tier 1	Inspection	
GRA-MPU-020	An MPU shall report a normative service status message as defined by OMS.	Tier 1	Demonstration	
GRA-MPU-021	A generated MPU Status shall be available on the ASB Critical Abstraction Layer.	Tier 1	Demonstration	
GRA-MPU-022	An MPU shall use the MEL for all information exchanges between AMS GRA Services and MFAs.	Tier 1	Inspection	
GRA-MPU-024	An MPU that interacts with the CDS or Cryptographic Unit shall use the CDS IDD and CU IDD respectively for all exchanges with the Cryptographic Unit or CDS.	Tier 1	Inspection	
GRA-MPU-025	An MPU shall report faults with the associated fault status.	Tier 1	Inspection	Off-nominal behavior detailed in compliance documentation.
GRA-MPU-026	An MPU shall implement Required interfaces specified IAW the AMS GRA Model.	Tier 1	Inspection	
GRA-MPU-027	An MPU shall implement all interfaces with the level of Openness specified IAW the AMS GRA Model.	Tier 1	Analysis	
GRA-MPU-029	An MPU shall implement only the interfaces in the corresponding MPU interface table.	Tier 2	Inspection	
GRA-MPU-030	An MPU shall document interface openness IAW the Compliance Assessment Approach.	Tier 1	Inspection	
GRA-MPU-031	An MPU shall include a model compatible with MBSE content contained in the AMS GRA SK release with which the MPU is compliant.	Tier 2	Inspection	
GRA-MPU-032	An MPU shall include a model that identifies all external MPU interfaces.	Tier 2	Inspection	

Id	Text	Compli-ance Tier	Recommended Verification Method	Notes
GRA-MPU-039	An MPU shall provide a CLASSIFICATION.md file defining security classification of included file contents.	Tier 2	Inspection	
GRA-MPU-040	An MPU shall provide a DATARIGHTS.md file defining data rights of included file contents.	Tier 2	Inspection	
GRA-MPU-041	An MPU shall document SAST results for all microprocessor software source code in the MPU, IAW the MPU documentation template.	Tier 1	Inspection	
GRA-MPU-042	An MPU shall document DAST results for all microprocessor software in the MPU, IAW the MPU documentation template.	Tier 1	Inspection	
GRA-MPU-046	An MPU shall provide automated tests of non-COTS microprocessor software with at least 50% source code structural coverage (excluding third-party COTS or open source libraries).	Tier 1	Demonstration	Only applicable to MPUs that include non-COTS microprocessor code.
GRA-MPU-047	An MPU with microprocessor code (excluding third-party COTS or open source libraries) shall only use relative file paths in the execution environment.	Tier 2	Inspection	
GRA-MPU-048	An MPU with microprocessor code (excluding third-party COTS or open source libraries) shall only use relative file paths in the build environment.	Tier 2	Inspection	
GRA-MPU-051	An MPU shall include a software bill of materials in an open and non-proprietary format.	Tier 1	Inspection	
GRA-MPU-053	An MPU's DPs shall each contain content only from that MPU.	Tier 3	Inspection	
GRA-MPU-057	An MPU's DP that targets a GPP Processing Component shall consist of a single OCI-compliant container.	Tier 1	Demonstration	
GRA-MPU-058	An MPU shall encrypt classified DaR using NSA-approved algorithms.	Tier 1	Analysis	
GRA-MPU-059	An MPU that decrypts and/or encrypts classified data and/or CPI shall only store classified plain text keys in volatile memory.	Tier 1	Inspection	
GRA-MPU-066	An MPU shall support the use of a Mission System-provided RoT.	Tier 1	Demonstration	

Id	Text	Compli-ance Tier	Recommended Verification Method	Notes
GRA-MPU-069	An MPU shall format MEL data IAW the MEL IDD corresponding to the MFA Type. Open additions are permitted.	Tier 1	Demonstration	
GRA-MPU-070	An MPU using a MEL API Implementation to interact with an MFA shall use the interface elements defined by the Common MEL API corresponding to the MFA Type with Open Additions to that MEL.	Tier 1	Demonstration	
GRA-MPU-071	An MPU shall be built, packaged, and delivered to persistent storage via automated pipeline.	Tier 1	Inspection	
GRA-MPU-074	An MPU that processes or decrypts CPI on a built-for-export platform shall fully implement OMS Standard.	Tier 3	Analysis	
GRA-MPU-077	An MPU shall provide automated tests of non-COTS microprocessor software with at least 80% source code structural coverage (excluding third-party COTS or open source libraries).	Tier 3	Demonstration	
GRA-MPU-078	An MPU model shall be capable of automated verification with AMS GRA models.	Tier 3	Inspection	
GRA-MPU-079	An MPU shall include a model that provides definition of all external MPU interfaces.	Tier 3	Inspection	
GRA-MPU-080	An MPU shall format MEL data IAW the MEL IDD corresponding to the MFA Type.	Tier 2	Demonstration	
GRA-MPU-081	An MPU using a MEL API Implementation to interact with an MFA shall use only the interface elements defined by the Common MEL API corresponding to the MFA Type.	Tier 2	Demonstration	
GRA-MPU-082	An MPU shall pass 70% or more of applicable jobs in a AMS GRA compliant pipeline.	Tier 2	Inspection	
GRA-MPU-083	An MPU shall pass 100% of applicable jobs in a AMS GRA compliant pipeline.	Tier 3	Inspection	

Id	Text	Compli-ance Tier	Recommended Verification Method	Notes
GRA-MPU-085	An MPU shall report faults with the associated fault status using the applicable interface. (See notes).	Tier 2	Inspection	Cryptographic Unit MPUss and CDS MPUss report faults as defined in the CU IDD and CDS IDD respectively. Local Function MPUs report faults as defined in Local Function DK. MFA MPUs report faults as defined in the applicable MEL IDD. AMS GRA Service, CMM, DPP, Data Storage, Switch, OMS Service, and PCaB MPUs report faults using a software interface in UCI format.
GRA-MPU-086	An MPU shall use the CMM IDD for all information exchanges between the MPU and the CMM.	Tier 1	Inspection	
GRA-MPU-090	An MPU shall provide a configuration file named service.yml with contents in accordance with the Service MPU Configuration Template YAML file.	Tier 1	Inspection	

Table 3.0.1: An AMS GRA Service MPU's Adherence to the Compliance Requirements Determines Its Compliance Score. See the Compliance Assessment Approach Volume for More Details.

A.0 Acronyms

ABB	Avionics Backplane Bus
AMS	Agile Mission Suite
API	Application Programming Interface
ASB	Abstract Service Bus
CCF	Compute Cluster Farm
CDS	Cross Domain Solution
CM	Countermeasure
CMM	Countermeasures Manager
COTS	Commercial-Off-The-Shelf
CPI	Critical Program Information
CU	Cryptographic Unit
DaR	Data at Rest
DAST	Dynamic Application Security Testing
DK	Development Kit
DP	Digital Payload
DPP	Digital Payload Processor
FPGA	Field-Programmable Gate Array
GPP	General Purpose Processor
GPU	Graphics Processing Unit
GRA	Government Reference Architecture
IAW	In Accordance With
IDD	Interface Description Document
IR	Infrared
MBSE	Model-Based Systems Engineering
MEL	MFA Encapsulation Layer
MFA	Multi-Function Aperture
MFP	Multi-Function Processor

MPU	Minimum Procurable Unit
MSI	Mission System Integrator
NSA	National Security Agency
OCI	Open Container Initiative
OMS	Open Mission Systems
PCaB	Processor Chassis and Backplane
RF	Radio Frequency
RoT	Root of Trust
SAST	Static Application Security Testing
SK	Starter Kit
SM	Security Module
UCI	Universal Command and Control Interface
VITA	VME bus International Trade Association
VME	Versa Module Eurocard
YAML	YAML Ain't Markup Language

B.0 Revision History

The Revision History table, **Table B.0.1**, captures the major changes made for a given AMS GRA volume as the Starter Kit (SK) continues to mature.

Revision	Description	Date	Notes
-	This revision adds a revision table to the volume.	May 22, 2024	SK11.0
A	This revision implements changes from AG-16510 into the document.	September 16, 2024	SK11.0
B	This revision adds prose updates and clarifications. GRA-MPU-003 and GRA-MPU-051 are updated to reflect current functionality and for clarity. Adds new requirement GRA-MPU-071.	April 24, 2026	AMS GRA 14.0

Table B.0.1: Revision History Captures the Major Changes Made to the AMS GRA Service MPU Volume Between Documentation Releases.